Math in Attention

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TODO: Might add some introductions later

Prove / Show

$$\underset{m\times n}{Y} = \underset{m\times k}{X}\ \underset{k\times n}{W}$$

$$dX = dY \cdot W^T$$

$$dW = X^T \cdot dY$$

$$\begin{cases} \dfrac{\partial L}{\partial X} = \sum_{ij} \dfrac{\partial L}{\partial Y_{ij}} \cdot \dfrac{\partial Y_{ij}}{\partial X} \\ dX = \sum_{ij} dY_{ij} \cdot \dfrac{\partial Y_{ij}}{\partial X} \end{cases}$$

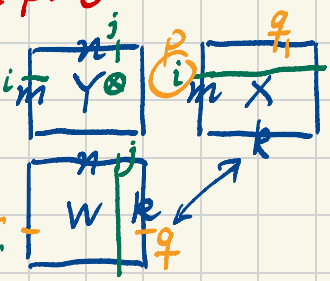
dX: $m\times k$; dY_{ij}: ①; $\dfrac{\partial Y_{ij}}{\partial X}$: $m\times k$

$$Y_{ij} = \sum_r X_{ir} \cdot W_{rj}$$

$$dX_{pq} = \sum_{ij} dY_{ij} \left(\frac{\partial Y_{ij}}{\partial X}\right)_{pq}$$

$$\begin{cases} i=p: \left(\dfrac{\partial Y_{pj}}{\partial X}\right)_{pq} \\ i\neq p: 0 \end{cases}$$

?



$$dX_{pq} = \sum_j dY_{pj} \cdot \left(\frac{\partial Y_{pj}}{\partial X}\right)_{pq}$$

$$= W_{qj}$$

$$dX_{pq} = \sum_j dY_{pj} \cdot (W^T)_{jp} = (dY \cdot W^T)_{pq}$$

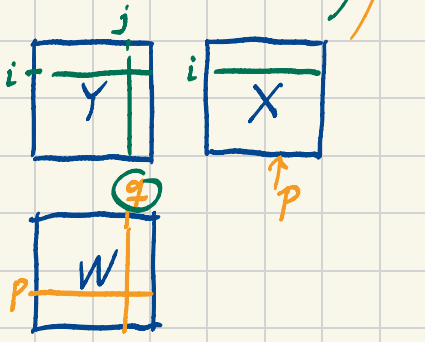
$$dX = dY \cdot W^T$$

similar:

$$\begin{cases} \dfrac{\partial L}{\partial W} = \sum_{ij} \dfrac{\partial L}{\partial Y_{ij}} \cdot \dfrac{\partial Y_{ij}}{\partial W} \\ dW_{pq} = \sum_{ij} dY_{ij} \cdot \left(\dfrac{\partial Y_{ij}}{\partial W}\right)_{pq} \end{cases}$$

$$\begin{cases} j=q: \left(\dfrac{\partial Y_{iq}}{\partial W}\right)_{pq} = X_{ip} \\ j\neq q: 0 \end{cases}$$

$$Y_{ij} = \sum_r X_{ir} \cdot W_{rj}$$



$$dW_{pq} = \sum_i dY_{iq} \cdot X_{ip}$$

$$= \sum_i (X^T)_{pi} \cdot dY_{iq}$$

reorder the inner product

$$= (X^T \cdot dY)_{pq} \quad \text{so} \quad dW = X^T \cdot dY$$

$P = \mathrm{softmax}(S)$ $dS = ?$ Solve.

P: $m \times n$, S: $m \times n$

$P_i = \mathrm{softmax}(S_i)$

already knows: $p = \mathrm{softmax}(s)$, p: $n \times 1$, s: $n \times 1$

$$\begin{cases} J = \mathrm{diag}(p) - p \cdot p^T \\ J_{ij} = p_i(\delta_{ij} - p_j) \end{cases}$$

$dp = J^T \cdot ds$

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

It's a bool expression widly used in Sums

$$\begin{cases} \frac{\partial L}{\partial S} = \sum_{ij} \frac{\partial L}{\partial P_{ij}} \cdot \frac{\partial P_{ij}}{\partial S} \\ dS = \sum_{ij} dP_{ij} \cdot \frac{\partial P_{ij}}{\partial S} \end{cases}$$

dS: $m \times L$, $\frac{\partial P_{ij}}{\partial S}$: $m \times L$

$$dS_{pq} = \sum_{ij} dP_{ij} \cdot \left(\frac{\partial P_{ij}}{\partial S}\right)_{pq} \quad \begin{cases} i = p: \left(\frac{\partial P_{pj}}{\partial S}\right)_{pq} \\ i \neq p: \ 0 \end{cases}$$

$$dS_{pq} = \sum_{j} dP_{pj} \cdot \left(\frac{\partial P_{pj}}{\partial S}\right)_{pq} = (J_p)_{jq}$$

$$\begin{cases} P_{pj} = \frac{e^{S_{pj}}}{Z_p} \\ Z_p = \sum_{k=1}^{L} e^{S_{pk}} \end{cases}$$

$$= P_{pj}(\delta_{jq} - P_{pq})$$

$$dS_{pq} = \sum_j dP_{pj} \cdot P_{pj} \cdot (\delta_{jq} - P_{pq})$$

really useful. now we don't have $\sum_{j \neq q}$

$$dS_{pq} = dP_{pq} \cdot P_{pq} - P_{pq} \cdot \sum_j dP_{pj} \cdot P_{pj}$$

$$dS_{pq} = P_{pq} \cdot (dP_{pq} - \underbrace{\sum_j dP_{pj} \cdot P_{pj}}_{\text{a dot product}})$$

$$dS_{pq} = P_{pq} \cdot (dP_{pq} - \langle dP_p, P_p \rangle)$$

broadcast on columns.

$$dS = P \odot (dP - dP \odot P \cdot 1_n \cdot 1_n^T)$$



still $\odot$ operations

code:

```
C = (P * dP).sum(axis = 1)
dS = P*(dP - C) # auto broadcast
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